

ECE 3337: Signals & Systems Analysis

Lecture 11: Laplace Transform

**Reference: Chapter 3
(Sections 3.2 – 3.3)**

Please turn off your gadgets now

Math Moment

Logarithms and Exponentials

$$f(x) = e^{ax}$$

$$\log f(x) = ax$$

$$e^{ax} e^{bx} = e^{(a+b)x}$$

$$\frac{e^{ax}}{e^{bx}} = e^{(a-b)x}$$

$$e^0 = 1$$

$$e^{jx} = \cos x + j \sin x$$

Limits

$$\lim_{x \rightarrow \infty} e^{-x} = 0$$

$$\lim_{x \rightarrow \infty} e^x = \infty$$

$$\lim_{x \rightarrow \infty} e^{jx} = ??$$

$$\lim_{x \rightarrow \infty} e^{(a+jb)x} = ??$$



Recap: The Laplace Transform

The Laplace transform is both general and powerful

$$\mathbf{X(s)} = \int\limits_{0-}^{+\infty} x(t)e^{-st} dt = \mathcal{L}[x(t)]$$

- **It smoothly handles a wide variety of signals, both continuous and discontinuous, and initial conditions**
- **It provides easy algebraic solutions, avoiding complex integrations**
- **We will restrict attention to causal signals and systems in this course that are described by the one-sided Laplace Transform**
 - **We can use tables of transforms instead of calculating complex integrals, for most signals of interest**



Recap: Laplace Transform Tables

Table 3-2: Examples of Laplace transform pairs. Note that $x(t) = 0$ for $t < 0^-$ and $T \geq 0$.

Laplace Transform Pairs		
	$x(t)$	$X(s) = \mathcal{L}[x(t)]$
1	$\delta(t)$	1
1a	$\delta(t - T)$	e^{-Ts}
2	$u(t)$	$\frac{1}{s}$
2a	$u(t - T)$	$\frac{e^{-Ts}}{s}$
3	$e^{-at} u(t)$	$\frac{1}{s + a}$
3a	$e^{-a(t-T)} u(t - T)$	$\frac{e^{-Ts}}{s + a}$
4	$t u(t)$	$\frac{1}{s^2}$
4a	$(t - T) u(t - T)$	$\frac{e^{-Ts}}{s^2}$
5	$t^2 u(t)$	$\frac{2}{s^3}$
6	$t e^{-at} u(t)$	$\frac{1}{(s + a)^2}$

7	$t^2 e^{-at} u(t)$	$\frac{2}{(s + a)^3}$
8	$t^{n-1} e^{-at} u(t)$	$\frac{(n-1)!}{(s + a)^n}$
9	$\sin(\omega_0 t) u(t)$	$\frac{\omega_0}{s^2 + \omega_0^2}$
10	$\sin(\omega_0 t + \theta) u(t)$	$\frac{s \sin \theta + \omega_0 \cos \theta}{s^2 + \omega_0^2}$
11	$\cos(\omega_0 t) u(t)$	$\frac{s}{s^2 + \omega_0^2}$
12	$\cos(\omega_0 t + \theta) u(t)$	$\frac{s \cos \theta - \omega_0 \sin \theta}{s^2 + \omega_0^2}$
13	$e^{-at} \sin(\omega_0 t) u(t)$	$\frac{\omega_0}{(s + a)^2 + \omega_0^2}$
14	$e^{-at} \cos(\omega_0 t) u(t)$	$\frac{s + a}{(s + a)^2 + \omega_0^2}$
15	$2e^{-at} \cos(bt - \theta) u(t)$	$\frac{e^{j\theta}}{s + a + jb} + \frac{e^{-j\theta}}{s + a - jb}$
15a	$e^{-at} \cos(bt - \theta) u(t)$	$\frac{(s + a) \cos \theta + b \sin \theta}{(s + a)^2 + b^2}$
16	$\frac{2t^{n-1}}{(n-1)!} e^{-at} \cos(bt - \theta) u(t)$	$\frac{e^{j\theta}}{(s + a + jb)^n} + \frac{e^{-j\theta}}{(s + a - jb)^n}$

Memorize the basic transforms and use the properties to derive the rest



Plan for Today

We will learn the basic properties of the Laplace transform

- We need these properties to handle complicated signals that are not directly listed in the L.T. tables
- We will often need to use a combination of these properties.

Table 3-1: Properties of the Laplace transform for causal functions; i.e., $x(t) = 0$ for $t < 0$.

Property	$x(t)$		$\mathbf{X(s)} = \mathcal{L}[x(t)]$
1. Multiplication by constant	$K x(t)$	$\longleftrightarrow$	$K \mathbf{X(s)}$
2. Linearity	$K_1 x_1(t) + K_2 x_2(t)$	$\longleftrightarrow$	$K_1 \mathbf{X_1(s)} + K_2 \mathbf{X_2(s)}$
3. Time scaling	$x(at), \quad a > 0$	$\longleftrightarrow$	$\frac{1}{a} \mathbf{X\left(\frac{s}{a}\right)}$
4. Time shift	$x(t - T) u(t - T)$	$\longleftrightarrow$	$e^{-Ts} \mathbf{X(s)}, \quad T \geq 0$
5. Frequency shift	$e^{-at} x(t)$	$\longleftrightarrow$	$\mathbf{X(s + a)}$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	$\longleftrightarrow$	$s \mathbf{X(s)} - x(0^-)$
7. Time 2nd derivative	$x'' = \frac{d^2x}{dt^2}$	$\longleftrightarrow$	$s^2 \mathbf{X(s)} - sx(0^-) - x'(0^-)$
8. Time integral	$\int_{0^-}^t x(t') dt'$	$\longleftrightarrow$	$\frac{1}{s} \mathbf{X(s)}$
9. Frequency derivative	$t x(t)$	$\longleftrightarrow$	$-\frac{d}{ds} \mathbf{X(s)} = -\mathbf{X'(s)}$
10. Frequency integral	$\frac{x(t)}{t}$	$\longleftrightarrow$	$\int_s^\infty \mathbf{X(s')} ds'$
11. Initial value	$x(0^+)$	$=$	$\lim_{s \rightarrow \infty} s \mathbf{X(s)}$
12. Final value	$\lim_{t \rightarrow \infty} x(t) = x(\infty)$	$=$	$\lim_{s \rightarrow 0} s \mathbf{X(s)}$
13. Convolution	$x_1(t) * x_2(t)$	$\longleftrightarrow$	$\mathbf{X_1(s)} \mathbf{X_2(s)}$



Example

Table 3-2: Examples of Laplace transform pairs. Note that $x(t) = 0$ for $t < 0^-$ and $T \geq 0$.

Laplace Transform Pairs		
	$x(t)$	$X(s) = \mathcal{L}[x(t)]$
1	$\delta(t)$	$\longleftrightarrow 1$
1a	$\delta(t - T)$	$\longleftrightarrow e^{-Ts}$
2	$u(t)$	$\longleftrightarrow \frac{1}{s}$
2a	$u(t - T)$	$\longleftrightarrow \frac{e^{-Ts}}{s}$
3	$e^{-at} u(t)$	$\longleftrightarrow \frac{1}{s + a}$
3a	$e^{-a(t-T)} u(t - T)$	$\longleftrightarrow \frac{e^{-Ts}}{s + a}$
4	$t u(t)$	$\longleftrightarrow \frac{1}{s^2}$
4a	$(t - T) u(t - T)$	$\longleftrightarrow \frac{e^{-Ts}}{s^2}$
5	$t^2 u(t)$	$\longleftrightarrow \frac{2}{s^3}$
6	$t e^{-at} u(t)$	$\longleftrightarrow \frac{1}{(s + a)^2}$

Example:

Find the inverse Laplace

Transform of $X(s) = \frac{1}{(s+2)^3}$

Looking up the table from right to left, we see that the inverse

L.T of $\frac{1}{s^3}$ is $\frac{1}{2} t^2 u(t)$. **ROW 5**

Next, we apply a frequency shift of 2 to obtain the answer:

$$x(t) = \frac{1}{2} e^{-2t} t^2 u(t)$$



Time Scaling Property

$$\mathcal{L}[x(at)] = ?$$

Note that when $a < 0$, the signal is no longer causal,
so, we restrict ourselves to $a > 0$

$$\mathcal{L}[x(at)] = \int_{0-}^{+\infty} x(at) e^{-st} dt$$

Let $u = at \Rightarrow du = a dt$ **Change of variable**

$$\mathcal{L}[x(at)] = \frac{1}{a} \int_{0-}^{+\infty} x(u) e^{-\left(\frac{s}{a}\right)u} du = \frac{1}{a} \mathbf{X}\left(\frac{s}{a}\right) \quad a > 0$$



Interpretation of Time Scaling

$$\mathcal{L}[x(at)] = \frac{1}{a} \mathbf{X}\left(\frac{s}{a}\right) \quad a > 0$$

- Time compression of a signal results in an expansion of its Laplace transform, and *vice versa (opposite behaviors)*
- Intuitively, faster-changing signals are associated with higher frequencies, and *vice versa*



Time Shift Property of the Laplace Transform

Delay by T

$$\mathcal{L}[x(t-T)u(t-T)] = \int_{0-}^{\infty} x(t-T)u(t-T)e^{-st} dt$$

Change of variable

$$\mu = t - T$$

$$= \int_{-T}^{\infty} x(\mu)u(\mu)e^{-s(\mu+T)} d\mu$$

The “delayer”

$$= \underline{e^{-sT}} \int_{-T}^{\infty} x(\mu)u(\mu)e^{-s\mu} d\mu$$

This only works for $T > 0$ since the signal may not be causal for a negative T

$$= \underline{e^{-sT}} \int_{0-}^{\infty} x(\mu)e^{-s\mu} d\mu = \underline{e^{-sT}} \mathbf{X}(s)$$



Example

Find the following Laplace transform:

$$\mathcal{L}[\sin(\omega_0(t - T))u(t - T)], \quad T \geq 0$$

Either from memory, or the table, we start with:

$$\mathcal{L}[\sin(\omega_0 t)u(t)] = \frac{\omega_0}{s^2 + \omega_0^2}$$

Multiplying by “the delayer” term gives us the answer:

$$\mathcal{L}[\sin(\omega_0(t - T))u(t - T)] = e^{-sT} \frac{\omega_0}{s^2 + \omega_0^2}$$



Example

Find the inverse Laplace transform of $X(s) = \frac{e^{-2s}}{(s+1)}$

We first find the inverse transform of $\frac{1}{(s+1)}$, which is $e^{-t}u(t)$

Next, we apply the delay of 2:

$$\Rightarrow x(t) = e^{-(t-2)}u(t-2)$$



Example

Find the Laplace transform of $x(t) = e^{-t}u(t - 1)$

This can be rewritten as $x(t) = e^{-1}e^{-(t-1)}u(t - 1)$

$$\Rightarrow \mathbf{X(s)} = e^{-1} \cdot \mathcal{L}[e^{-(t-1)}u(t - 1)]$$

$$= e^{-1} \cdot \frac{e^{-s}}{(s + 1)} = \frac{e^{-(s+1)}}{(s + 1)}$$



Frequency Shift

$$e^{-at}x(t) \leftrightarrow \mathbf{X(s + a)} \quad \text{The shift is in the s domain}$$

Easy to see that:

$$\begin{aligned}\mathcal{L}[e^{-at}x(t)] &= \int_{0-}^{\infty} e^{-at}x(t)e^{-st} dt \\ &= \int_{0-}^{\infty} e^{-(s+a)t}x(t) dt = \mathbf{X(s + a)}\end{aligned}$$

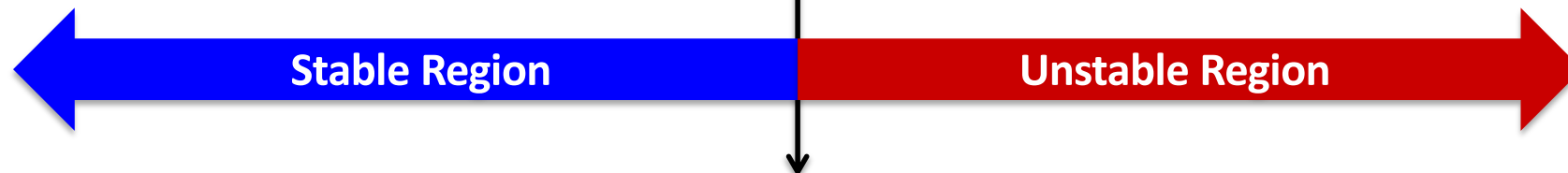
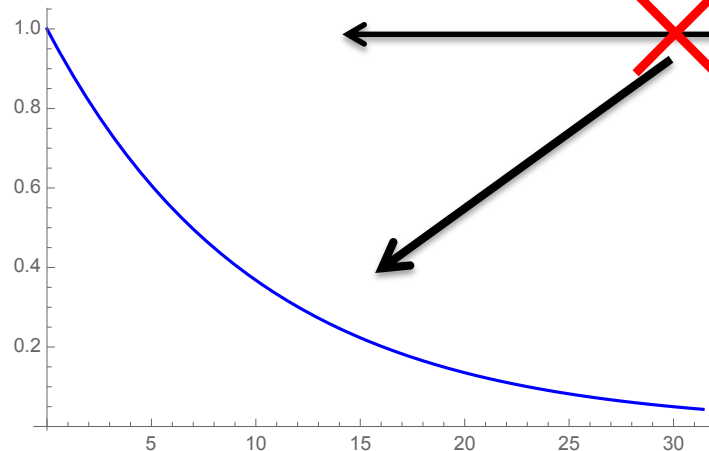
Basically, replace all instances of s with $(s + a)$



Example

$$e^{-at}u(t), a > 0 \leftrightarrow \frac{1}{(s+a)}$$

Pole at $s = -a$

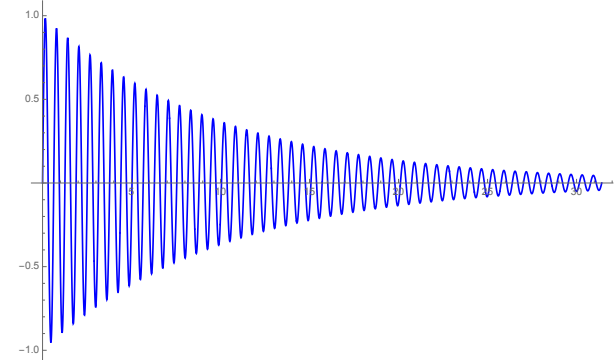




Example

Find the Laplace transform of the exponentially **damped sinusoid**:

$$\mathcal{L}[\underbrace{e^{-at}}_{\text{green}} \underbrace{\sin(\omega_0 t)u(t)}_{\text{blue}}], \quad a > 0$$



This pattern occurs frequently because we often “**dampen**” various **signals** using an exponential multiplier from an RC circuit

We already know that $\mathcal{L}[\sin(\omega_0 t)u(t)] = \frac{\omega_0}{s^2 + \omega_0^2}$
So,

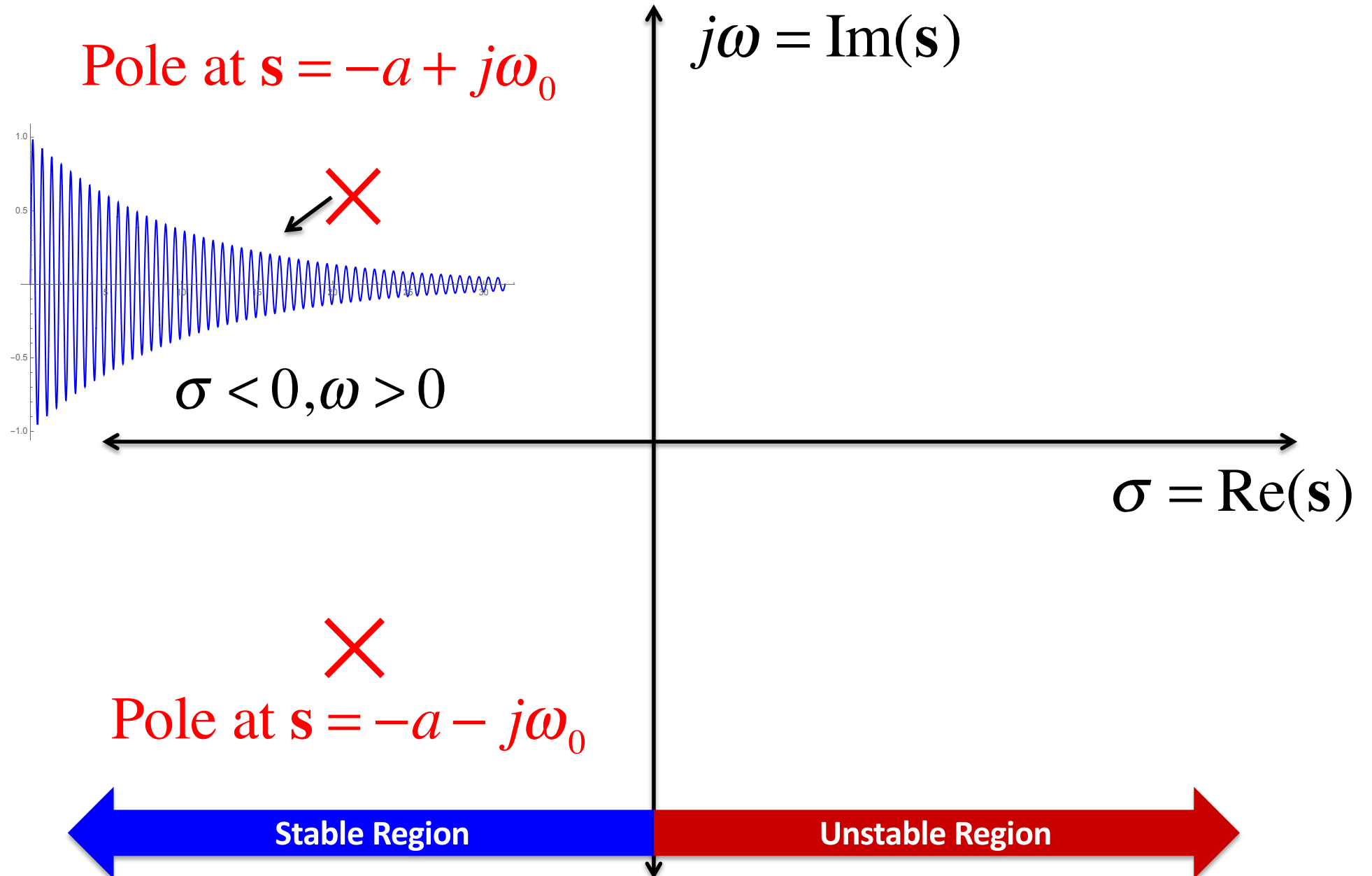
$$\mathcal{L}[e^{-at} \sin(\omega_0 t)u(t)] = \frac{\omega_0}{(s + a)^2 + \omega_0^2}$$



Example (contd.)

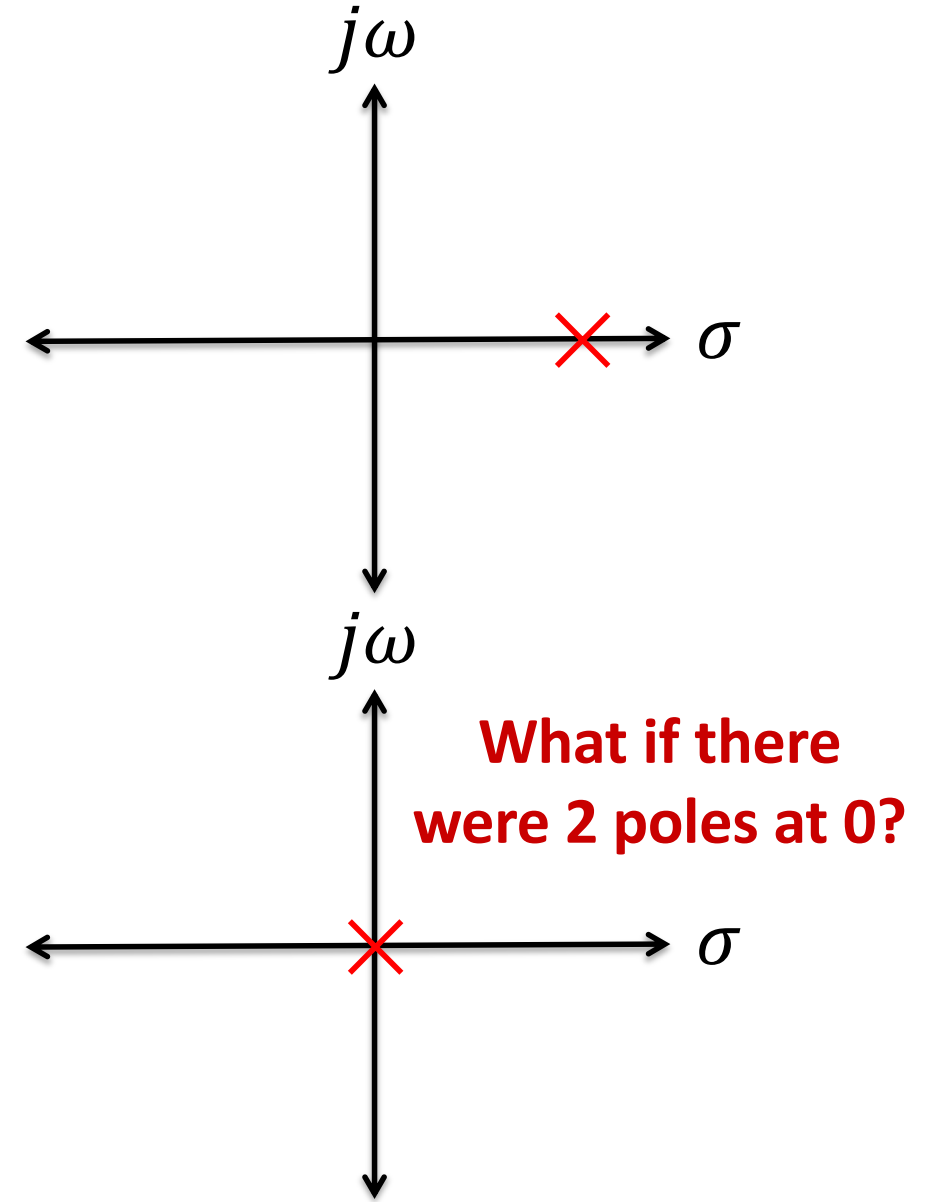
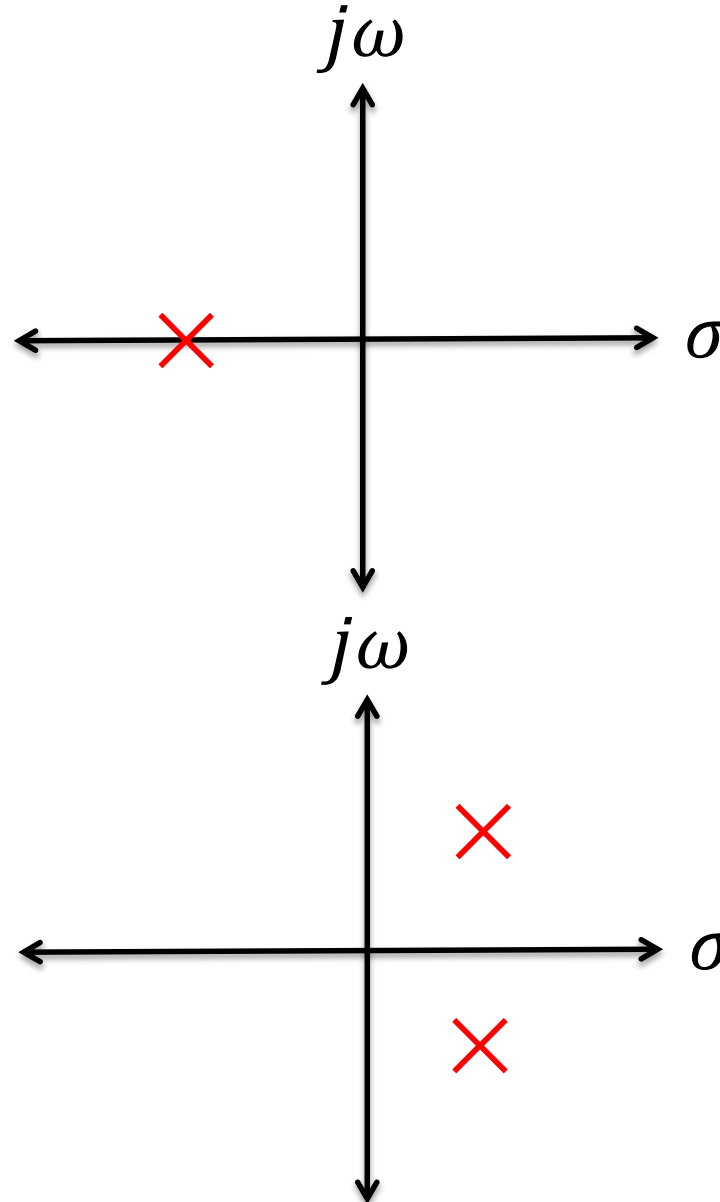
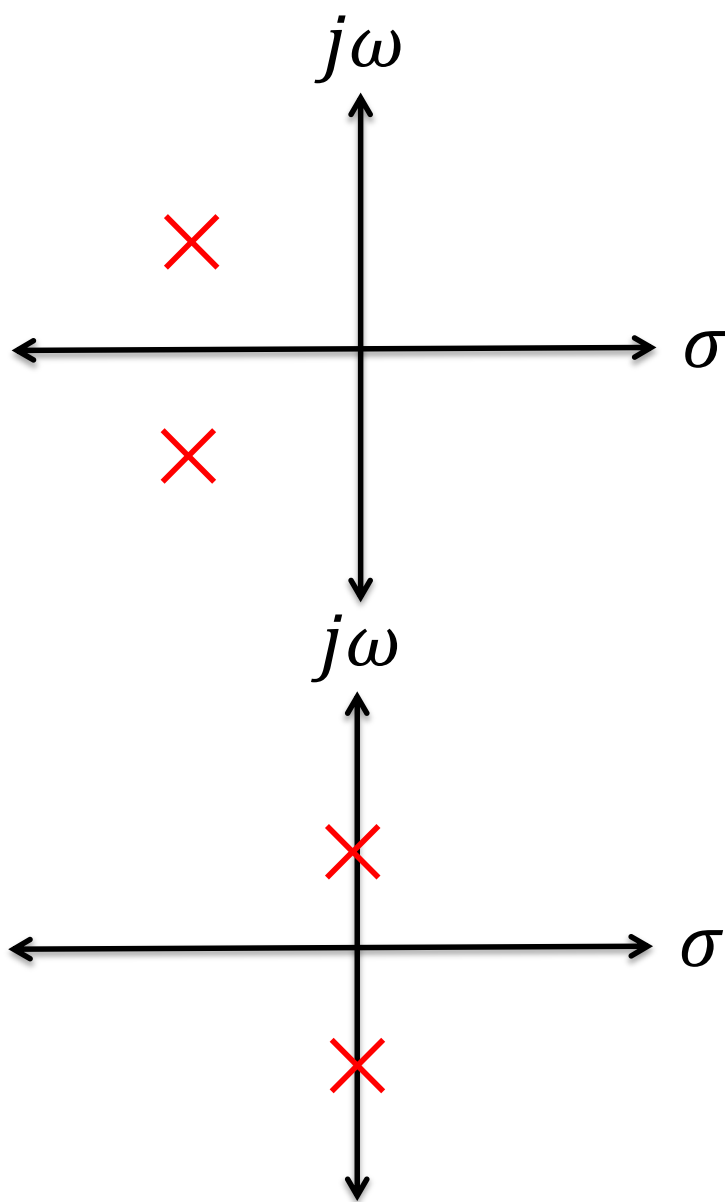
$$e^{-at} \sin(\omega_0 t) u(t)$$

$a > 0$





What Can We Say About the Following Systems?





Signal Differentiation

We differentiate a signal when we need to detect changes, or measure the rate of change

$$\mathcal{L} \left[\frac{dx(t)}{dt} \right] = \int_{0-}^{\infty} \underbrace{\frac{dx(t)}{dt}}_{\text{red bracket}} \underbrace{e^{-st}}_{\text{green bracket}} dt$$

This integral consists of a product of two functions

To solve it, we must do “integration by parts”



Signal Differentiation

$$\mathcal{L}\left[\frac{dx(t)}{dt}\right] = \int_{0-}^{\infty} \underbrace{\frac{dx(t)}{dt}}_g \underbrace{e^{-st}}_f dt$$

Integrating by parts:

$$\begin{aligned} &= x(t)e^{-st} \Big|_{0-}^{\infty} - \int_{0-}^{\infty} x(t)(-s)e^{-st} dt \\ &= -x(0-) + s \int_{0-}^{\infty} x(t)e^{-st} dt \end{aligned}$$

$$\mathcal{L}\left[\frac{dx(t)}{dt}\right] = -x(0-) + s\mathbf{X}(s)$$

**Basically, differentiating a signal multiplies its Laplace transform by s
Remember to subtract the initial condition**



Repeated Differentiation

The differentiation property can be applied repeatedly to calculate Laplace transforms of higher derivatives

$$\begin{aligned}\mathcal{L}\left[\frac{d^2x(t)}{dt^2}\right] &= \mathbf{s}[\mathbf{sX}(\mathbf{s}) - x(0-)] - x'(0-) \\ &= \mathbf{s}^2\mathbf{X}(\mathbf{s}) - \mathbf{s}x(0-) - x'(0-)\end{aligned}$$

General Formula for n^{th} derivative:

$$\mathcal{L}\left[\frac{d^nx(t)}{dt^n}\right] = \mathbf{s}^n\mathbf{X}(\mathbf{s}) - \mathbf{s}^{n-1}x(0-) - \mathbf{s}^{n-2}x'(0-) \dots - \mathbf{s}^0x^{(n-1)}(0-)$$



Example

Transform the following differential equation into the Laplace domain:

$$\underbrace{\frac{d^2 y(t)}{dt^2}} + \underbrace{\frac{dy(t)}{dt}} + \underbrace{y(t)} = x(t)$$

$$\underbrace{\{s^2 Y(s) - sy(0-) - y'(0-)\}} + \underbrace{\{sY(s) - y(0-)\}} + \underbrace{\{Y(s)\}} = X(s)$$

$$(s^2 + s + 1)Y(s) - sy(0-) - y'(0-) - y(0-) = X(s)$$

When the initial conditions are all zero, this simplifies to:

$$(s^2 + s + 1)Y(s) = X(s)$$



Connecting Time and Frequency

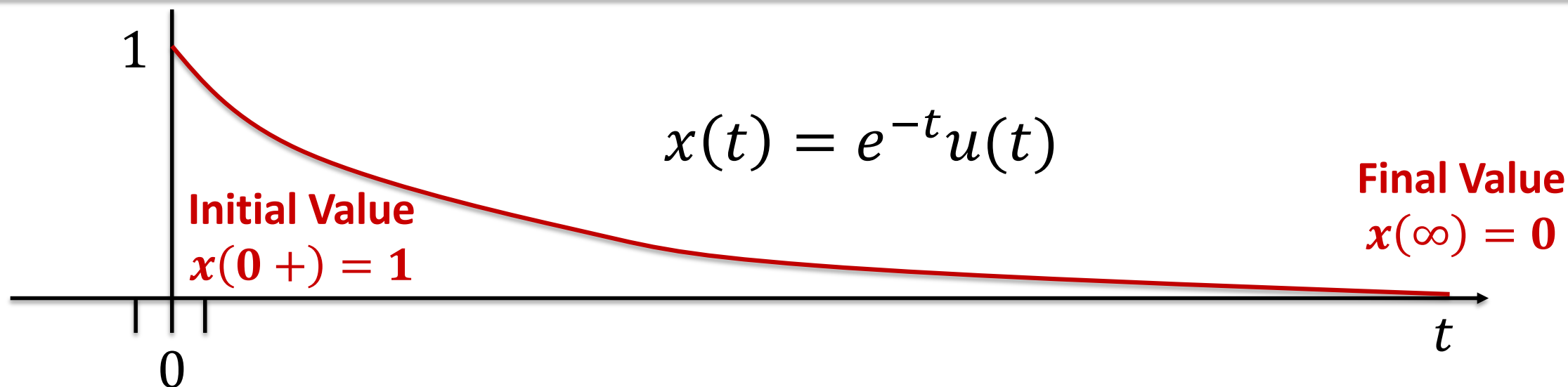
$$\mathcal{L}\left[\frac{dx(t)}{dt}\right] = -x(0^-) + sX(s)$$
A large red arrow pointing upwards from the text below to the term $x(0^-)$ in the equation above.

The differentiation property is fascinating because it connects the frequency domain to a specific moment in the time domain $t = 0$ –

Can we exploit this connection?



Special Moments in the Life of a Causal Signal

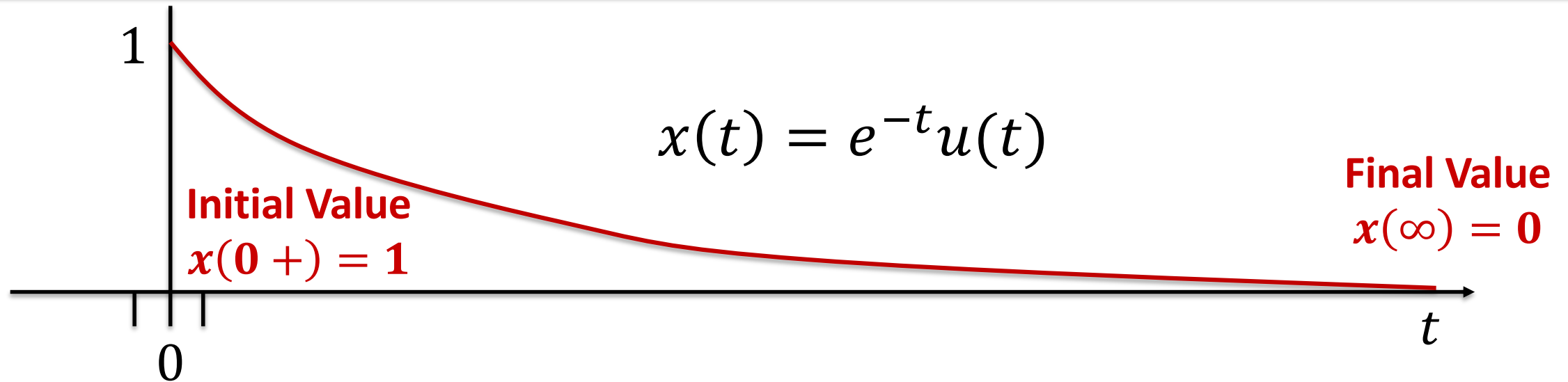


The initial and final values of a signal/system are especially interesting:

- The initial value $x(0^+)$ usually results from the initial conditions at $t = 0^-$ – and any impulse-like inputs at $t < 0^+$
- The final value $x(\infty)$ is when the system has (hopefully) “settled”



Special Moments in the Life of a Causal Signal



The extreme time-domain values $t \rightarrow 0$ and $t \rightarrow \infty$ correspond to the opposite extreme frequency-domain values

$t \rightarrow 0$ corresponds to $s \rightarrow \infty$

$t \rightarrow \infty$ corresponds to $s \rightarrow 0$



Extreme Case #1: $s \rightarrow 0, t \rightarrow \infty$

$$\int_{0-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt = \mathbf{sX(s)} - x(0-) \text{ Differentiation Property}$$

$$\Rightarrow \lim_{s \rightarrow 0} \left\{ \int_{0-}^{\infty} \frac{dx(t)}{dt} e^{-\cancel{st}^1} dt \right\} = \lim_{s \rightarrow 0} \{ \mathbf{sX(s)} \} - x(0-)$$

$$\int_{0-}^{\infty} \frac{dx(t)}{dt} dt = \lim_{s \rightarrow 0} \{ \mathbf{sX(s)} \} - x(0-)$$

$$x(\infty) - x(0-) = \lim_{s \rightarrow 0} \{ \mathbf{sX(s)} \} - x(0-)$$

$$\Rightarrow \lim_{s \rightarrow 0} \{ \mathbf{sX(s)} \} = x(\infty)$$

**This is called the
final value theorem**



Example

Find the final value of the signal $x(t) = e^{-t}u(t)$

Time domain: $x(\infty) = \lim_{t \rightarrow \infty} \{e^{-t}u(t)\} = 0$

Frequency domain: $X(s) = \frac{1}{s+1}$

$$x(\infty) = \lim_{s \rightarrow 0} \{sX(s)\} = \lim_{s \rightarrow 0} \left\{ \frac{s}{s+1} \right\} = 0$$

Same Answer



Example

Find the final value of the signal:

$$\mathbf{X(s)} = \frac{25s(s+3)}{(s+1)(s^2+2s+36)}$$

$$x(\infty) = \lim_{s \rightarrow 0} \{s\mathbf{X(s)}\}$$

$$= \lim_{s \rightarrow 0} \frac{25s^2(s+3)}{(s+1)(s^2+2s+36)}$$

$$= 0$$

Using this theorem, we can calculate the final (settling) value without explicitly calculating the inverse $x(t)$



Example

Find the final value of the following signal:

$$X(s) = \frac{s}{s^2 + \omega_0^2}$$

If we directly apply the theorem we get:

$$x(\infty) = \lim_{s \rightarrow 0} \{sX(s)\} = \lim_{s \rightarrow 0} \left\{ \frac{s^2}{s^2 + \omega_0^2} \right\} = 0$$

Now, why is this answer suspect?



Caution About Final Value Theorem

For our example,

$$x(t) = \cos(\omega_0 t)u(t)$$

As $t \rightarrow \infty$, this undamped oscillatory signal does not converge to a unique value

Moral: don't apply this theorem blindly.

Check to see if $sX(s)$ has undamped oscillations (roots on the $(j\omega)$ axis), has unstable (poles in the right half plane), or has more than one root at the origin (e.g., $1/s^2$ corresponds to a ramp that grows indefinitely).



Another Misleading Example

Find the final value of the following signal:

$$X(s) = \frac{s + 2}{s(s - 1)}$$

If we apply the theorem we get:

$$\lim_{s \rightarrow 0} \{sX(s)\} = \lim_{s \rightarrow 0} \left\{ \frac{s + 2}{s - 1} \right\} \rightarrow -2$$

This limit is misleading since this system is BIBO unstable

$$(x(t) = -2u(t) + 3e^t u(t))$$



Pole in the right half plane



Yet Another Example

Find the final value of the following signal:

$$X(s) = \frac{s^2 + 3s + 2}{s^2(s^2 + \omega_0^2)}$$

If we apply the theorem we get:

$$\lim_{s \rightarrow 0} \{sX(s)\} = \lim_{s \rightarrow 0} \left\{ \frac{s^2 + 3s + 2}{s(s^2 + \omega_0^2)} \right\} \rightarrow \infty$$

Systems with more than one root at the origin are also problematic. We cannot use this theorem



Extreme Case #2: $s \rightarrow \infty, t \rightarrow 0$

$$\int_{0-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt = \mathbf{sX(s)} - x(0-) \quad \text{Differentiation Property}$$

$$\Rightarrow \lim_{s \rightarrow \infty} \left\{ \int_{0-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt \right\} = \lim_{s \rightarrow \infty} \left\{ \int_{0-}^{0+} \frac{dx(t)}{dt} \underbrace{e^{-st}}_{\text{Becomes 1 as } t \rightarrow 0} dt \right\} + \underbrace{\lim_{s \rightarrow \infty} \left\{ \int_{0+}^{\infty} \frac{dx(t)}{dt} e^{-st} dt \right\}}_{\text{Becomes 0 as } s \rightarrow \infty}$$

$$= x(0+) - x(0-)$$

$$= \lim_{s \rightarrow \infty} \{\mathbf{sX(s)}\} - x(0-)$$

$$\Rightarrow \lim_{s \rightarrow \infty} \{\mathbf{sX(s)}\} = x(0+)$$

This result is called the
initial value theorem



Example

Find the initial value of the signal $X(s) = \frac{1}{s+1}$

$$\lim_{s \rightarrow \infty} \{sX(s)\} = \lim_{s \rightarrow \infty} \left\{ \frac{s}{s+1} \right\} = 1$$

**This is correct since $x(t) = e^{-t}u(t)$
and this tends to 1 as $t \rightarrow 0$**



Example

Find the initial value of the following signal:

$$X(s) = \frac{25s(s+3)}{(s+1)(s^2+2s+36)}$$

$$x(0+) = \lim_{s \rightarrow \infty} \{sX(s)\} = \lim_{s \rightarrow \infty} \frac{25s^2(s+3)}{(s+1)(s^2+2s+36)}$$

$$= \lim_{s \rightarrow \infty} \frac{25 \left(1 + \frac{3}{s}\right)}{\left(1 + \frac{1}{s}\right) \left(1 + \frac{2}{s} + \frac{36}{s^2}\right)} = 25$$

Using this theorem, we can calculate the initial value without explicitly calculating the inverse $x(t)$



Example

Find the initial value of the following signal:

$$\mathbf{X(s)} = \frac{s}{s + 1} \qquad x(t) = \delta(t) - e^{-t}u(t)$$

$$x(0+) = \lim_{s \rightarrow \infty} \{s\mathbf{X(s)}\} = \lim_{s \rightarrow \infty} \left\{ \frac{s^2}{s + 1} \right\} = \infty$$

- The initial value theorem is usable when the $s\mathbf{X(s)}$ denominator's order is greater than the numerator's, so a unique limit exists as $s \rightarrow \infty$
- Interestingly, the locations of the poles do not matter for the Initial Value Theorem in the way they did for the Final Value Theorem



Time Integration

$$\mathcal{L}\left[\int_{0^-}^t x(\tau) d\tau\right] = ?$$

$$\text{Let } g(t) = \int_{0^-}^t x(\tau) d\tau \quad \text{and} \quad g(0^-) = 0$$

Now if $g(t) \leftrightarrow \mathbf{G}(s)$, then

$$\mathbf{X}(s) = \mathcal{L}\left[\frac{dg(t)}{dt}\right] = s\mathbf{G}(s) - g(0^-) = s\mathbf{G}(s)$$

Therefore,

$$\mathbf{G}(s) = \frac{\mathbf{X}(s)}{s}$$

$$\int_{0^-}^t x(\tau) d\tau \leftrightarrow \frac{\mathbf{X}(s)}{s}$$

Basically, integrating a signal divides its Laplace transform by s

Remember that this does not work when $s = 0$



Time Integration

If the lower limit of the integration is $-\infty$, we have the following corollary:

$$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{\mathbf{X}(\mathbf{s})}{\mathbf{s}} + \frac{1}{\mathbf{s}} \int_{-\infty}^{0-} x(\tau) d\tau$$

This version is needed occasionally for dealing with initial conditions



Example

We can use the integration property to figure out the following Laplace transform relationships at a glance!

$$\delta(t) \leftrightarrow 1 \quad \text{Impulse}$$

$$u(t) \leftrightarrow \frac{1}{s} \quad \text{Step}$$

$$r(t) \leftrightarrow \frac{1}{s^2} \quad \text{Ramp}$$



Differentiation in the Frequency Domain

$$\mathbf{X}(s) = \int_{0-}^{+\infty} x(t)e^{-st} dt = \mathcal{L}[x(t)]$$

Let's differentiate with respect to s on both sides:

$$\begin{aligned}\frac{d}{ds}[\mathbf{X}(s)] &= \int_{0-}^{+\infty} \frac{d}{ds}[x(t)e^{-st}] dt \\ &= \int_{0-}^{+\infty} -t x(t)e^{-st} dt \\ &= \mathcal{L}[-t x(t)]\end{aligned}$$

Multiplying by $-t$ in the time domain is equivalent to differentiating $\mathbf{X}(s)$ in the frequency domain



Example

Find the following Laplace transform:

$$\mathcal{L}\left[\underbrace{t e^{-at} u(t)}\right]$$

This is another classic pattern we must learn to recognize.

We already know that:

$$\mathcal{L}\left[e^{-at} u(t)\right] = \frac{1}{s + a}$$

So,

$$\mathcal{L}\left[t e^{-at} u(t)\right] = \downarrow \frac{d}{ds} \left[\frac{1}{s + a} \right] = \frac{1}{(s + a)^2}$$



Frequency Domain Integration

$$\frac{x(t)}{t} \leftrightarrow \int_s^{\infty} \mathbf{X}(s') ds'$$

Dividing by t in the time domain is equivalent to integrating $\mathbf{X}(s)$ in the frequency domain
This property is the “dual cousin” of the frequency-domain differentiation property



Combined Example

Let us consider a more complicated example that combines many of the ideas we have been discussing. Calculate the following:

$$\mathcal{L}\left[t^2 e^{-3t} \cos(4t) u(t)\right]$$

When faced with a problem like this, we should first identify the helpful patterns from Tables 3-1 & 3-2:

$\cos(\omega_0 t) u(t)$: Known Laplace transform

e^{-3t} : Frequency Shift

t^2 : Frequency differentiation done twice



Combined Example

From Table 3-2: Our starting point is

$$\mathcal{L}[\cos(4t)u(t)] = \frac{s}{s^2 + 4^2}$$

Frequency Shift Property: $e^{-at}x(t) \leftrightarrow \mathbf{X}(s + a)$

$$\Rightarrow \mathcal{L}[e^{-3t} \cos(4t)u(t)] = \frac{s + 3}{(s + 3)^2 + 4^2}$$



Combined Example

$$\mathcal{L}\left[e^{-3t} \cos(4t)u(t)\right] = \frac{s+3}{(s+3)^2 + 4^2}$$

Frequency Differentiation: $\mathcal{L}[t x(t)] = -\frac{d}{ds}[\mathbf{X}(s)]$

Done Twice: $\mathcal{L}[t^2 x(t)] = \frac{d^2}{ds^2}[\mathbf{X}(s)]$

$$\Rightarrow \mathcal{L}\left[t^2 e^{-3t} \cos(4t)u(t)\right] = \frac{d^2}{ds^2} \left\{ \frac{s+3}{(s+3)^2 + 4^2} \right\}$$



Summary

Today, we studied basic Laplace transform Properties

– Super useful!

They are summarized in Table 3-1.

We use Tables 3-1 and 3-2 together to calculate the forward and inverse transforms of signals

We will learn later that circuit elements will transform cleanly to the **s** domain

– Makes it easier to do all our work in the Laplace domain!

Table 3-1: Properties of the Laplace transform for causal functions; i.e., $x(t) = 0$ for $t < 0$.

Property	$x(t)$		$\mathbf{X}(s) = \mathcal{L}[x(t)]$
1. Multiplication by constant	$K x(t)$	$\longleftrightarrow$	$K \mathbf{X}(s)$
2. Linearity	$K_1 x_1(t) + K_2 x_2(t)$	$\longleftrightarrow$	$K_1 \mathbf{X}_1(s) + K_2 \mathbf{X}_2(s)$
3. Time scaling	$x(at), \quad a > 0$	$\longleftrightarrow$	$\frac{1}{a} \mathbf{X}\left(\frac{s}{a}\right)$
4. Time shift	$x(t - T) u(t - T)$	$\longleftrightarrow$	$e^{-Ts} \mathbf{X}(s), \quad T \geq 0$
5. Frequency shift	$e^{-at} x(t)$	$\longleftrightarrow$	$\mathbf{X}(s + a)$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	$\longleftrightarrow$	$s \mathbf{X}(s) - x(0^-)$
7. Time 2nd derivative	$x'' = \frac{d^2x}{dt^2}$	$\longleftrightarrow$	$s^2 \mathbf{X}(s) - sx(0^-) - x'(0^-)$
8. Time integral	$\int_{0^-}^t x(t') dt'$	$\longleftrightarrow$	$\frac{1}{s} \mathbf{X}(s)$
9. Frequency derivative	$t x(t)$	$\longleftrightarrow$	$-\frac{d}{ds} \mathbf{X}(s) = -\mathbf{X}'(s)$
10. Frequency integral	$\frac{x(t)}{t}$	$\longleftrightarrow$	$\int_s^\infty \mathbf{X}(s') ds'$
11. Initial value	$x(0^+)$	$=$	$\lim_{s \rightarrow \infty} s \mathbf{X}(s)$
12. Final value	$\lim_{t \rightarrow \infty} x(t) = x(\infty)$	$=$	$\lim_{s \rightarrow 0} s \mathbf{X}(s)$
13. Convolution	$x_1(t) * x_2(t)$	$\longleftrightarrow$	$\mathbf{X}_1(s) \mathbf{X}_2(s)$



For the Next Class

Work through the textbook examples in Sections 3.3 – 3.4

Study & Smartly Memorize Tables 3.1 and 3.2 : *We will use these tables for daily problem solving*

For the next class, please study Sections 3-4, 3-5

How to Study:

- 1. Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts**
- 2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
- 3. Write down your questions and bring them to class**